BTS SIO SISR · SERVICES RÉSEAU LINUX

Service DNS sous Linux

Fiche procédure — BIND9 serveur principal et secondaire, zones directe et inverses, intégration DHCP KEA, routeur Linux et pfSense

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Sommaire

1. Architecture et plan d'adressage
2. SRV-DEB2 — serveur DNS principal (BIND9)
3. Zones de résolution inverse
4. DHCP, RTUX, SRV-DEB1 et pfSense — utilisation du nouveau DNS
5. Client — tests de résolution
6. DNS secondaire sur SRV-DEB1
7. Basculement — panne du DNS principal
8. Mise à jour dynamique DHCP → DNS
9. Synthèse

Architecture et plan d'adressage

Trois réseaux reliés par un routeur Linux (RTUX), pfSense vers Internet. LAN Serveurs passé en 172.20.30.0/28 pour ce TP.

0.1 — RTUX — trois interfaces et routage

ens3 vers pfSense (LAN-RT), ens4 vers LAN Serveurs (nouveau segment 172.20.30.0/28), ens5 vers LAN Clients. ip_forward = 1 : RTUX route entre les trois réseaux.

sisr@RTUX

```
sisr@RTUX:~$ hostname; ip -4 -br a | grep -v lo; ip r; cat /proc/sys/net/ipv4/ip_forward
RTUX
ens3             UP             172.20.20.6/29
ens4             UP             172.20.30.14/28
ens5             UP             172.18.127.254/17
default via 172.20.20.2 dev ens3 proto static
172.18.0.0/17 dev ens5 proto kernel scope link src 172.18.127.254
172.20.20.0/29 dev ens3 proto kernel scope link src 172.20.20.6
172.20.30.0/28 dev ens4 proto kernel scope link src 172.20.30.14
1
```

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0.2 — SRV-DEB1 et SRV-DEB2 — adressage sur LAN Serveurs

Passerelle des serveurs : RTUX (172.20.30.14).

thomas@pop-os

```
thomas@pop-os:~$ for h in 172.20.30.1 172.20.30.2; do ssh -o StrictHostKeyChecking=no -o UserKnownHostsFil
e=/dev/null -o LogLevel=ERROR sisr@$h 'echo "## $(hostname)"; ip -4 -br a | grep -v lo; ip r | grep defaul
t'; done
## SRV-DEB1
ens3             UP             172.20.30.1/28
default via 172.20.30.14 dev ens3 proto static
## SRV-DEB2
ens3             UP             172.20.30.2/28
default via 172.20.30.14 dev ens3 proto static
```

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0.3 — Client DEB-CLT — bail DHCP reçu via le relais

Avant le TP, le DHCP distribue encore 1.1.1.1 comme DNS.

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ hostname; ip -4 -br a | grep -v lo; ip r | grep default; cat /etc/resolv.conf | grep -v '^
#'
DEB-CLT
ens3             UP             172.18.0.1/17
default via 172.18.127.254 dev ens3 proto dhcp src 172.18.0.1 metric 100

nameserver 1.1.1.1
search hiron.th
```

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SRV-DEB2 — serveur DNS principal (BIND9)

SRV-DEB2 héberge la zone hiron.th en maître : redirecteur, zone directe, enregistrements.

1.1 — Installation de BIND9 sur SRV-DEB2

Paquets : bind9 (serveur, service named), bind9-utils (named-checkconf, named-checkzone), dnsutils (dig, nslookup).

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ dpkg -l bind9 bind9-utils dnsutils | grep ^ii | awk '{print $1, $2, $3}'; systemctl statu
s named --no-pager | head -12
ii bind9 1:9.18.49-1~deb12u3
ii bind9-utils 1:9.18.49-1~deb12u3
ii dnsutils 1:9.18.49-1~deb12u3
● named.service - BIND Domain Name Server
     Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 09:48:34 CET; 1min 46s ago
       Docs: man:named(8)
   Main PID: 1046 (named)
     Status: "running"
      Tasks: 5 (limit: 1141)
     Memory: 30.7M
        CPU: 35ms
     CGroup: /system.slice/named.service
             └─1046 /usr/sbin/named -f -u bind
```

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1.2 — Fichiers de configuration de BIND

named.conf.options = réglages globaux ; named.conf.local = déclaration des zones ; db.* = fichiers de zone.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ ls -l /etc/bind
total 48
-rw-r--r-- 1 root root 2928 13 oct.  21:49 bind.keys
-rw-r--r-- 1 root root  255 13 oct.  21:41 db.0
-rw-r--r-- 1 root root  271 13 oct.  21:41 db.127
-rw-r--r-- 1 root root  237 13 oct.  21:41 db.255
-rw-r--r-- 1 root root  353 13 oct.  21:41 db.empty
-rw-r--r-- 1 root root  270 13 oct.  21:41 db.local
-rw-r--r-- 1 root bind  458 13 oct.  21:41 named.conf
-rw-r--r-- 1 root bind  498 13 oct.  21:41 named.conf.default-zones
-rw-r--r-- 1 root bind  165 13 oct.  21:41 named.conf.local
-rw-r--r-- 1 root bind  846 13 oct.  21:41 named.conf.options
-rw-r----- 1 bind bind  100 10 nov.  09:48 rndc.key
-rw-r--r-- 1 root root 1317 13 oct.  21:41 zones.rfc1918
```

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1.3 — named.conf.options — redirecteur vers 1.1.1.1

Le redirecteur transmet à 1.1.1.1 toute requête hors de la zone locale ; seuls les réseaux du TP (acl reseaux_tp) peuvent interroger le serveur.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ cat /etc/bind/named.conf.options; sudo named-checkconf && echo 'named-checkconf : OK (auc
une erreur)'
acl "reseaux_tp" {
        127.0.0.1;
        172.20.30.0/28;     // LAN Serveurs
        172.18.0.0/17;      // LAN Clients
        172.20.20.0/29;     // LAN RTUX / pfSense
};

options {
        directory "/var/cache/bind";

        forwarders {
                1.1.1.1;
        };
        forward only;

        recursion yes;
        allow-query { reseaux_tp; };
        allow-recursion { reseaux_tp; };

        dnssec-validation auto;
        listen-on { any; };
        listen-on-v6 { none; };
};
named-checkconf : OK (aucune erreur)
```

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1.4 — Test du redirecteur avec dig

La 2e requête est servie depuis le cache de BIND (temps proche de 0 ms).

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ systemctl is-active named; dig @127.0.0.1 debian.org | grep -E 'status|ANSWER SECTION|IN
\s+A|Query time|SERVER'; echo '--- 2e requête (cache) ---'; dig @127.0.0.1 debian.org | grep -E 'Query tim
e'
active
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33861
;debian.org.                    IN      A
;; ANSWER SECTION:
debian.org.             67      IN      A       151.101.130.132
debian.org.             67      IN      A       151.101.194.132
debian.org.             67      IN      A       151.101.2.132
debian.org.             67      IN      A       151.101.66.132
;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
--- 2e requête (cache) ---
;; Query time: 0 msec
```

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1.5 — Déclaration de la zone hiron.th (named.conf.local)

type master : SRV-DEB2 détient la version originale de la zone.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ grep -v '^//' /etc/bind/named.conf.local | sed '/^\s*$/d'; sudo named-checkconf && echo
'named-checkconf : OK'
zone "hiron.th" {
        type master;
        file "/etc/bind/db.hiron.th";
};
named-checkconf : OK
```

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1.6 — Fichier de zone db.hiron.th

Enregistrements A des 2 serveurs, des 3 interfaces de RTUX et de la carte LAN de pfSense ; alias www vers SRV-DEB1.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ cat /etc/bind/db.hiron.th; sudo named-checkzone hiron.th /etc/bind/db.hiron.th
$TTL 86400
@       IN      SOA     srv-deb2.hiron.th. admin.hiron.th. (
                        2026111001      ; Serial (AAAAMMJJnn)
                        3600            ; Refresh
                        900             ; Retry
                        604800          ; Expire
                        86400 )         ; TTL négatif

; Serveur de noms faisant autorité
@               IN      NS      srv-deb2.hiron.th.

; Serveurs
srv-deb1        IN      A       172.20.30.1
srv-deb2        IN      A       172.20.30.2

; RTUX : une entrée par interface
rtux-rt         IN      A       172.20.20.6
rtux-srv        IN      A       172.20.30.14
rtux-clt        IN      A       172.18.127.254

; pfSense (carte LAN)
pfsense         IN      A       172.20.20.2

; Alias
www             IN      CNAME   srv-deb1
zone hiron.th/IN: loaded serial 2026111001
OK
```

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1.7 — Test de la zone avec dig

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ systemctl is-active named; dig @127.0.0.1 www.hiron.th +noall +answer; for n in srv-deb1 srv-deb2 rtux-rt rtux-srv rtux-clt pfsense; do dig @127.0.0.1 $n.hiron.th +short | sed "s/^/$n.hiron.th -> /"; done
active
www.hiron.th.           86400   IN      CNAME   srv-deb1.hiron.th.
srv-deb1.hiron.th.      86400   IN      A       172.20.30.1
srv-deb1.hiron.th -> 172.20.30.1
srv-deb2.hiron.th -> 172.20.30.2
rtux-rt.hiron.th -> 172.20.20.6
rtux-srv.hiron.th -> 172.20.30.14
rtux-clt.hiron.th -> 172.18.127.254
pfsense.hiron.th -> 172.20.20.2
```

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Zones de résolution inverse

Les zones in-addr.arpa permettent de retrouver un nom à partir d'une adresse IP (enregistrements PTR).

2.1 — Déclaration des zones inverses (named.conf.local)

Zone inverse = octets du réseau à l'envers + .in-addr.arpa. LAN Clients en /17 → zone du /16 englobant (18.172.in-addr.arpa).

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ grep -v '^//' /etc/bind/named.conf.local | sed '/^\s*$/d'; sudo named-checkconf && echo
'named-checkconf : OK'
zone "hiron.th" {
        type master;
        file "/etc/bind/db.hiron.th";
};
zone "30.20.172.in-addr.arpa" {
        type master;
        file "/etc/bind/db.172.20.30";
};
zone "18.172.in-addr.arpa" {
        type master;
        file "/etc/bind/db.172.18";
};
named-checkconf : OK
```

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2.2 — Zone inverse LAN Clients — PTR de RTUX

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ cat /etc/bind/db.172.18; sudo named-checkzone 18.172.in-addr.arpa /etc/bind/db.172.18; di
g -x 172.18.127.254 @127.0.0.1 +noall +answer
$TTL 86400
@       IN      SOA     srv-deb2.hiron.th. admin.hiron.th. (
                        2026111001 3600 900 604800 86400 )
@       IN      NS      srv-deb2.hiron.th.

254.127 IN      PTR     rtux-clt.hiron.th.
zone 18.172.in-addr.arpa/IN: loaded serial 2026111001
OK
254.127.18.172.in-addr.arpa. 86400 IN   PTR     rtux-clt.hiron.th.
```

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2.3 — Zone inverse LAN Serveurs — PTR de SRV-DEB1, SRV-DEB2 et RTUX

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ cat /etc/bind/db.172.20.30; sudo named-checkzone 30.20.172.in-addr.arpa /etc/bind/db.172.20.30; for ip in 172.20.30.1 172.20.30.2 172.20.30.14; do dig -x $ip @127.0.0.1 +noall +answer; done
$TTL 86400
@       IN      SOA     srv-deb2.hiron.th. admin.hiron.th. (
                        2026111001 3600 900 604800 86400 )
@       IN      NS      srv-deb2.hiron.th.

1       IN      PTR     srv-deb1.hiron.th.
2       IN      PTR     srv-deb2.hiron.th.
14      IN      PTR     rtux-srv.hiron.th.
zone 30.20.172.in-addr.arpa/IN: loaded serial 2026111001
OK
1.30.20.172.in-addr.arpa. 86400 IN      PTR     srv-deb1.hiron.th.
2.30.20.172.in-addr.arpa. 86400 IN      PTR     srv-deb2.hiron.th.
14.30.20.172.in-addr.arpa. 86400 IN     PTR     rtux-srv.hiron.th.
```

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DHCP, RTUX, SRV-DEB1 et pfSense — utilisation du nouveau DNS

Chaque machine et les clients DHCP doivent utiliser SRV-DEB2 comme serveur DNS.

3.1 — DHCP KEA — option DNS = SRV-DEB2 (sur les deux serveurs)

Les deux serveurs KEA (haute disponibilité) distribuent la même option domain-name-servers 172.20.30.2.

thomas@pop-os

```
thomas@pop-os:~$ for h in 172.20.30.2 172.20.30.1; do ssh -o StrictHostKeyChecking=no -o UserKnownHostsFil
e=/dev/null -o LogLevel=ERROR sisr@$h 'echo "## $(hostname)"; sudo grep -n domain-name /etc/kea/kea-dhcp4.
conf; sudo kea-dhcp4 -t /etc/kea/kea-dhcp4.conf >/dev/null 2>&1 && echo "kea-dhcp4 -t : configuration vali
de"; systemctl is-active kea-dhcp4-server'; done
## SRV-DEB2
28:        { "name": "domain-name", "data": "hiron.th" },
29:        { "name": "domain-name-servers", "data": "172.20.30.2" }
kea-dhcp4 -t : configuration valide
active
## SRV-DEB1
28:        { "name": "domain-name", "data": "hiron.th" },
29:        { "name": "domain-name-servers", "data": "172.20.30.2" }
kea-dhcp4 -t : configuration valide
active
```

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3.2 — SRV-DEB2 — /etc/hosts et résolution (resolv.conf via netplan)

/etc/resolv.conf est généré par systemd-resolved depuis netplan : le DNS se change donc dans netplan.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ grep -v '^#' /etc/hosts | sed '/^\s*$/d'; echo '--- netplan ---'; sudo grep -A4 nameserve
rs /etc/netplan/50-cloud-init.yaml; echo '--- /etc/resolv.conf ---'; grep -v '^#' /etc/resolv.conf | sed
'/^\s*$/d'; echo '--- tests ---'; dig srv-deb1.hiron.th +short; dig debian.org +short | head -2
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.20.30.2     srv-deb2.hiron.th       srv-deb2
--- netplan ---
            nameservers:
                addresses:
                - 172.20.30.2
                search:
                - hiron.th
--- /etc/resolv.conf ---
nameserver 172.20.30.2
search hiron.th
--- tests ---
172.20.30.1
151.101.2.132
151.101.130.132
```

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3.3 — SRV-DEB1 — /etc/hosts et DNS (netplan → resolv.conf)

Le DHCP de SRV-DEB1 distribue déjà 172.20.30.2 (étape précédente, configuration KEA identique sur les deux serveurs). getent passe par le résolveur système, comme une application.

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ grep -v '^#' /etc/hosts | sed '/^\s*$/d'; echo '--- netplan ---'; sudo grep -A4 nameserve
rs /etc/netplan/50-cloud-init.yaml; echo '--- /etc/resolv.conf ---'; grep -v '^#' /etc/resolv.conf | sed
'/^\s*$/d'; echo '--- tests (résolveur système) ---'; getent hosts www.hiron.th srv-deb2 pfsense; getent a
hostsv4 debian.org | head -1
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.20.30.1     srv-deb1.hiron.th       srv-deb1
--- netplan ---
            nameservers:
                addresses:
                - 172.20.30.2
                search:
                - hiron.th
--- /etc/resolv.conf ---
nameserver 172.20.30.2
search hiron.th
--- tests (résolveur système) ---
172.20.30.1     srv-deb1.hiron.th www.hiron.th
172.20.30.2     srv-deb2.hiron.th
172.20.20.2     pfsense.hiron.th
151.101.130.132 STREAM debian.org
```

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3.4 — RTUX — /etc/hosts et DNS (netplan → resolv.conf)

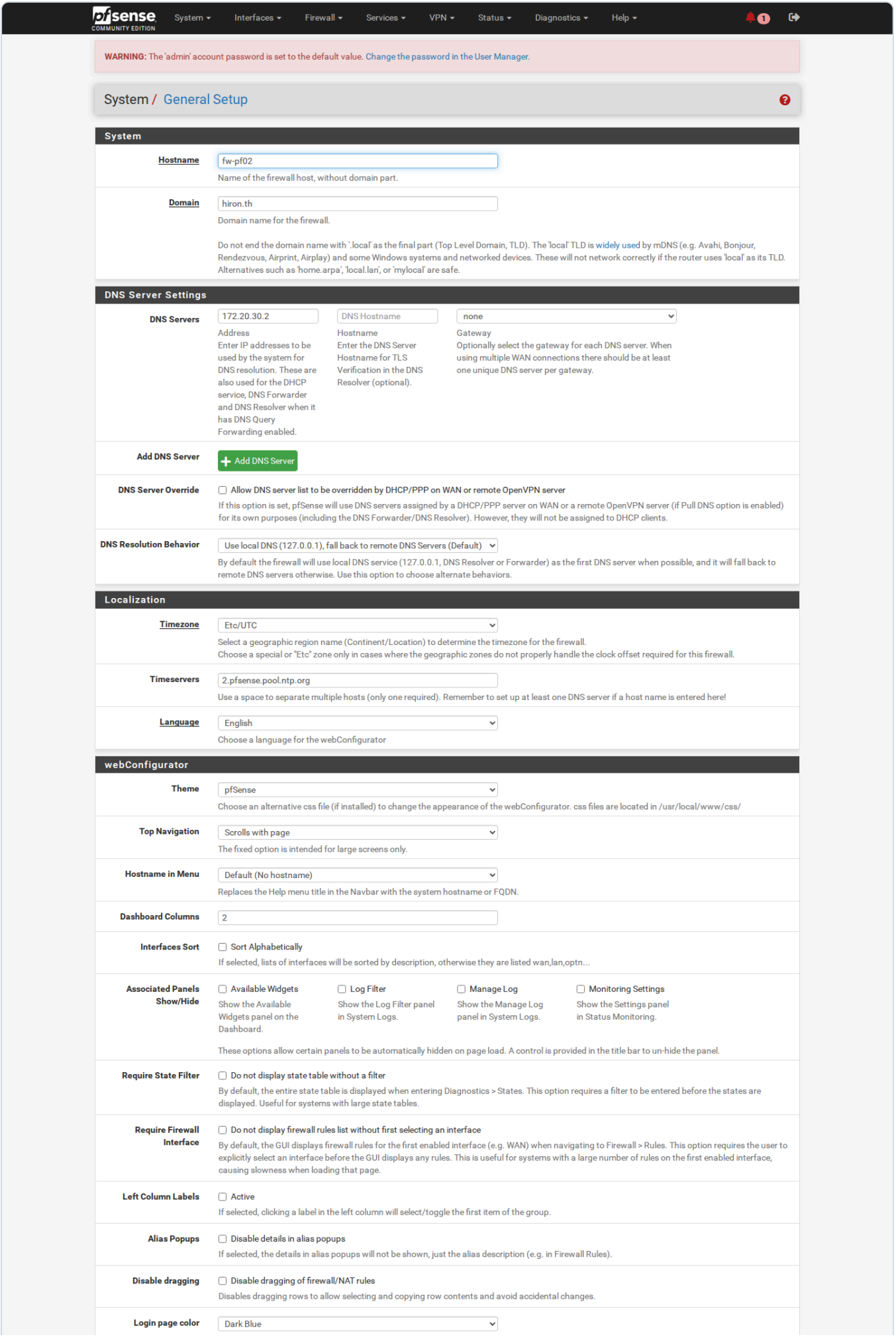
sisr@RTUX

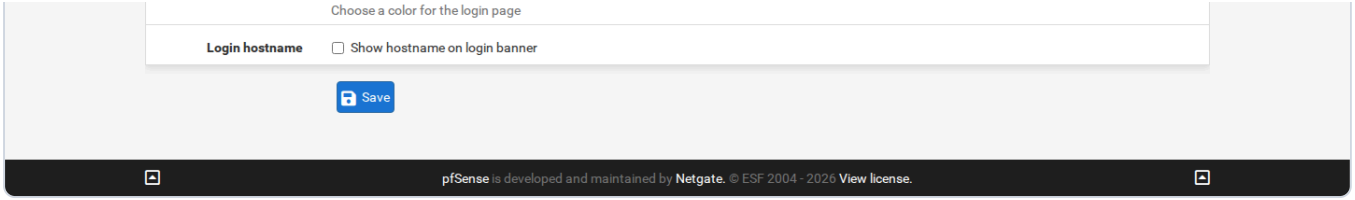
```
sisr@RTUX:~$ grep -v '^#' /etc/hosts | sed '/^\s*$/d'; echo '--- netplan ---'; sudo grep nameservers /etc/
netplan/60-lab.yaml; echo '--- /etc/resolv.conf ---'; grep -v '^#' /etc/resolv.conf | sed '/^\s*$/d'; echo
'--- tests (résolveur système) ---'; getent hosts srv-deb1 srv-deb2.hiron.th www; getent ahostsv4 debian.o
rg | head -1
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.20.20.6     rtux-rt.hiron.th        rtux-rt     rtux
172.20.30.14    rtux-srv.hiron.th       rtux-srv
172.18.127.254  rtux-clt.hiron.th       rtux-clt
--- netplan ---
      nameservers: {addresses: [172.20.30.2], search: [hiron.th]}
--- /etc/resolv.conf ---
nameserver 172.20.30.2
search hiron.th
--- tests (résolveur système) ---
172.20.30.1     srv-deb1.hiron.th
172.20.30.2     srv-deb2.hiron.th
172.20.30.1     srv-deb1.hiron.th www.hiron.th
151.101.2.132   STREAM debian.org
```

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3.5 — pfSense — System > General Setup : domaine et DNS

Domaine hiron.th, serveur DNS 172.20.30.2 (SRV-DEB2) ; la liste DNS n'est plus remplaçable par le DHCP du WAN.

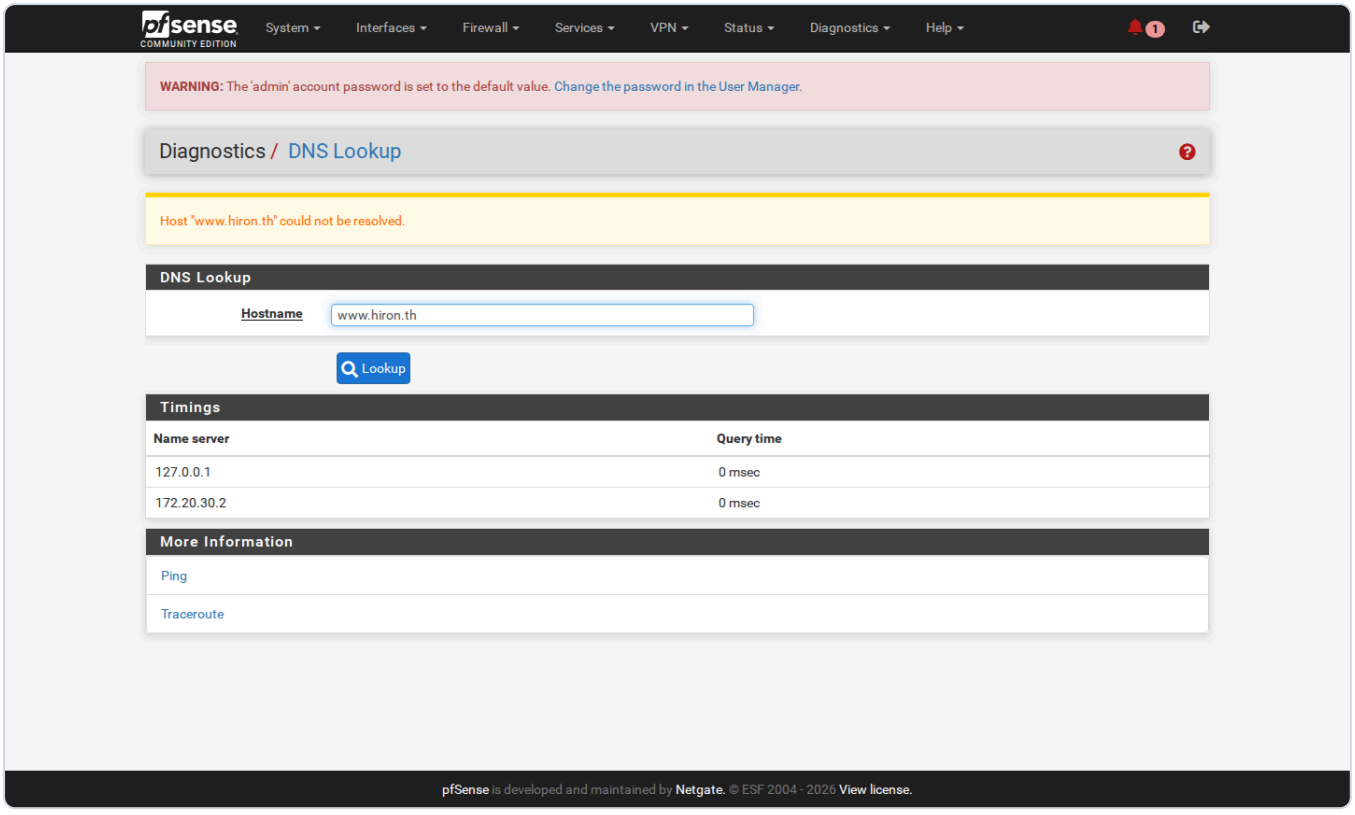

pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
System / General Setup
System
Hostname
fw-pf02
Name of the firewall host, without domain part.
Domain
hiron.th
Domain name for the firewall.
Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.
DNS Server Settings
DNS Servers
172.20.30.2
Address
Enter IP addresses to be used by the system for DNS resolution. These are also used for the DHCP service, DNS Forwarder and DNS Resolver when it has DNS Query Forwarding enabled.
DNS Hostname
Hostname
Enter the DNS Server Hostname for TLS Verification in the DNS Resolver (optional).
none
Gateway
Optionally select the gateway for each DNS server. When using multiple WAN connections there should be at least one unique DNS server per gateway.
Add DNS Server
Add DNS Server
DNS Server Override
Allow DNS server list to be overridden by DHCP/PPP on WAN or remote OpenVPN server
If this option is set, pfSense will use DNS servers assigned by a DHCP/PPP server on WAN or a remote OpenVPN server (if Pull DNS option is enabled) for its own purposes (including the DNS Forwarder/DNS Resolver). However, they will not be assigned to DHCP clients.
DNS Resolution Behavior
Use local DNS (127.0.0.1), fall back to remote DNS Servers (Default)
By default the firewall will use local DNS service (127.0.0.1, DNS Resolver or Forwarder) as the first DNS server when possible, and it will fall back to remote DNS servers otherwise. Use this option to choose alternate behaviors.
Localization
Timezone
Etc/UTC
Select a geographic region name (Continent/Location) to determine the timezone for the firewall.
Choose a special or "Etc" zone only in cases where the geographic zones do not properly handle the clock offset required for this firewall.
Timeservers
2.pfsense.pool.ntp.org
Use a space to separate multiple hosts (only one required). Remember to set up at least one DNS server if a host name is entered here!
Language
English
Choose a language for the webConfigurator
webConfigurator
Theme
pfSense
Choose an alternative css file (if installed) to change the appearance of the webConfigurator. css files are located in /usr/local/www/css/
Top Navigation
Scrolls with page
The fixed option is intended for large screens only.
Hostname in Menu
Default (No hostname)
Replaces the Help menu title in the Navbar with the system hostname or FQDN.
Dashboard Columns
2
Interfaces Sort
Sort Alphabetically
If selected, lists of interfaces will be sorted by description, otherwise they are listed wan,lan,optn...
Associated Panels Show/Hide
Available Widgets
Show the Available Widgets panel on the Dashboard.
Log Filter
Show the Log Filter panel in System Logs.
Manage Log
Show the Manage Log panel in System Logs.
Monitoring Settings
Show the Settings panel in Status Monitoring.
These options allow certain panels to be automatically hidden on page load. A control is provided in the title bar to un-hide the panel.
Require State Filter
Do not display state table without a filter
By default, the entire state table is displayed when entering Diagnostics > States. This option requires a filter to be entered before the states are displayed. Useful for systems with large state tables.
Require Firewall Interface
Do not display firewall rules list without first selecting an interface
By default, the GUI displays firewall rules for the first enabled interface (e.g. WAN) when navigating to Firewall > Rules. This option requires the user to explicitly select an interface before the GUI displays any rules. This is useful for systems with a large number of rules on the first enabled interface, causing slowness when loading that page.
Left Column Labels
Active
If selected, clicking a label in the left column will select/toggle the first item of the group.
Alias Popups
Disable details in alias popups
If selected, the details in alias popups will not be shown, just the alias description (e.g. in Firewall Rules).
Disable dragging
Disable dragging of firewall/NAT rules
Disables dragging rows to allow selecting and copying row contents and avoid accidental changes.
Login page color
Dark Blue



Source : https://172.20.20.2/system.php
Capturé le 10/11/2026 10:24:24

3.6 — pfSense — DNS Lookup en échec avant correction (incident)

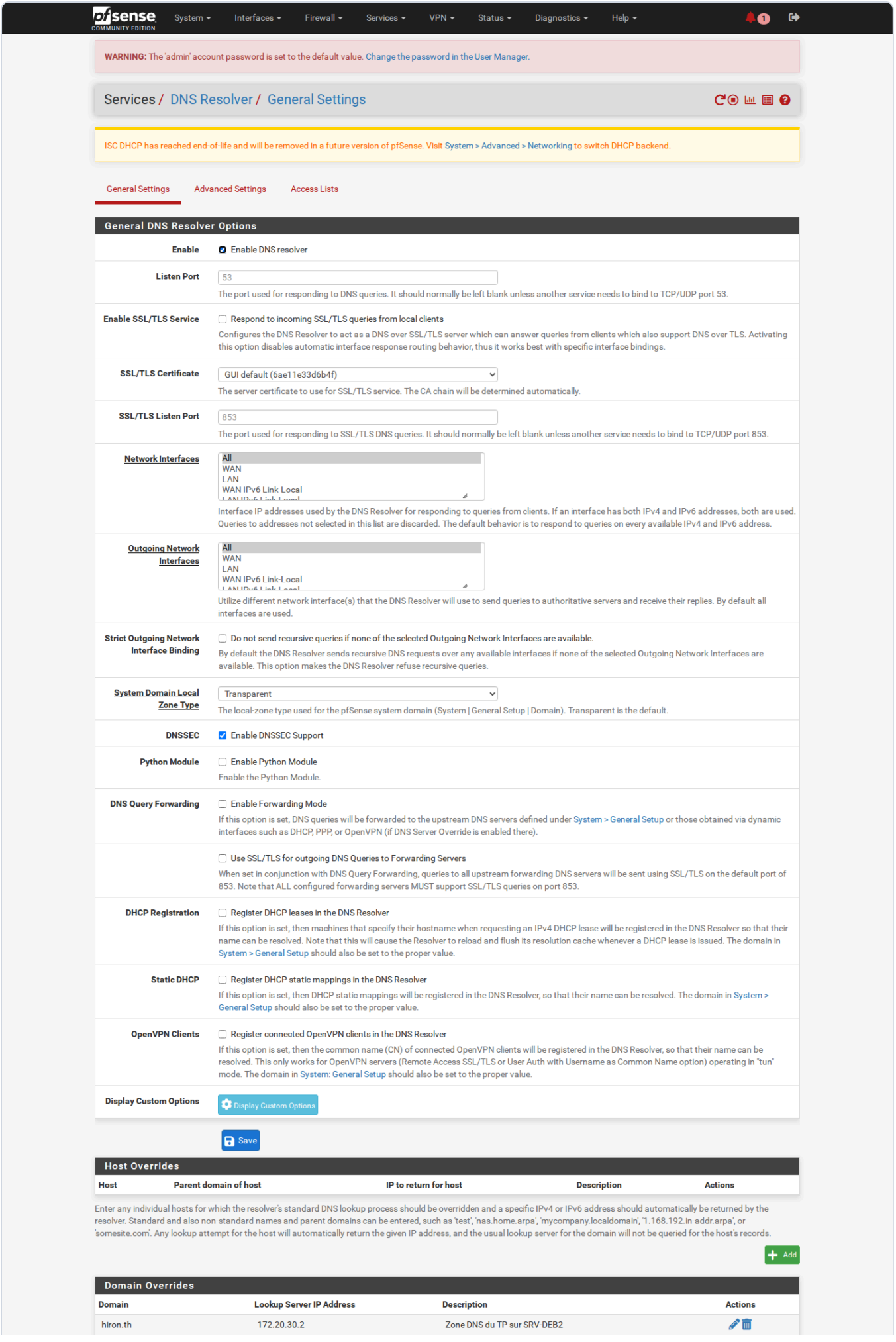
Incident : pfSense interroge d'abord son résolveur local Unbound (127.0.0.1), qui cherche hiron.th sur Internet et répond NXDOMAIN. NXDOMAIN étant une réponse valide, pfSense ne passe jamais à 172.20.30.2. Correction : Domain Overrides dans le DNS Resolver.

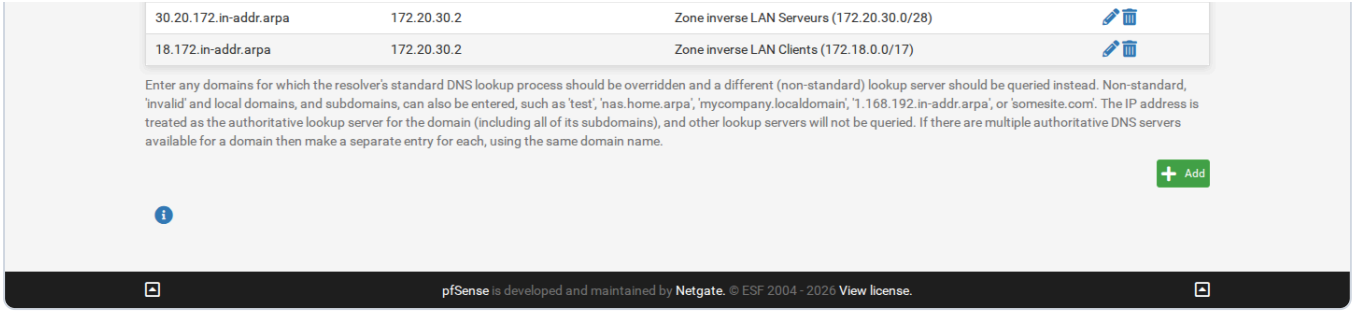


Source : https://172.20.20.2/diag_dns.php?host=www.hiron.th
Capturé le 10/11/2026 10:25:22

3.7 — pfSense — DNS Resolver : Domain Overrides vers SRV-DEB2

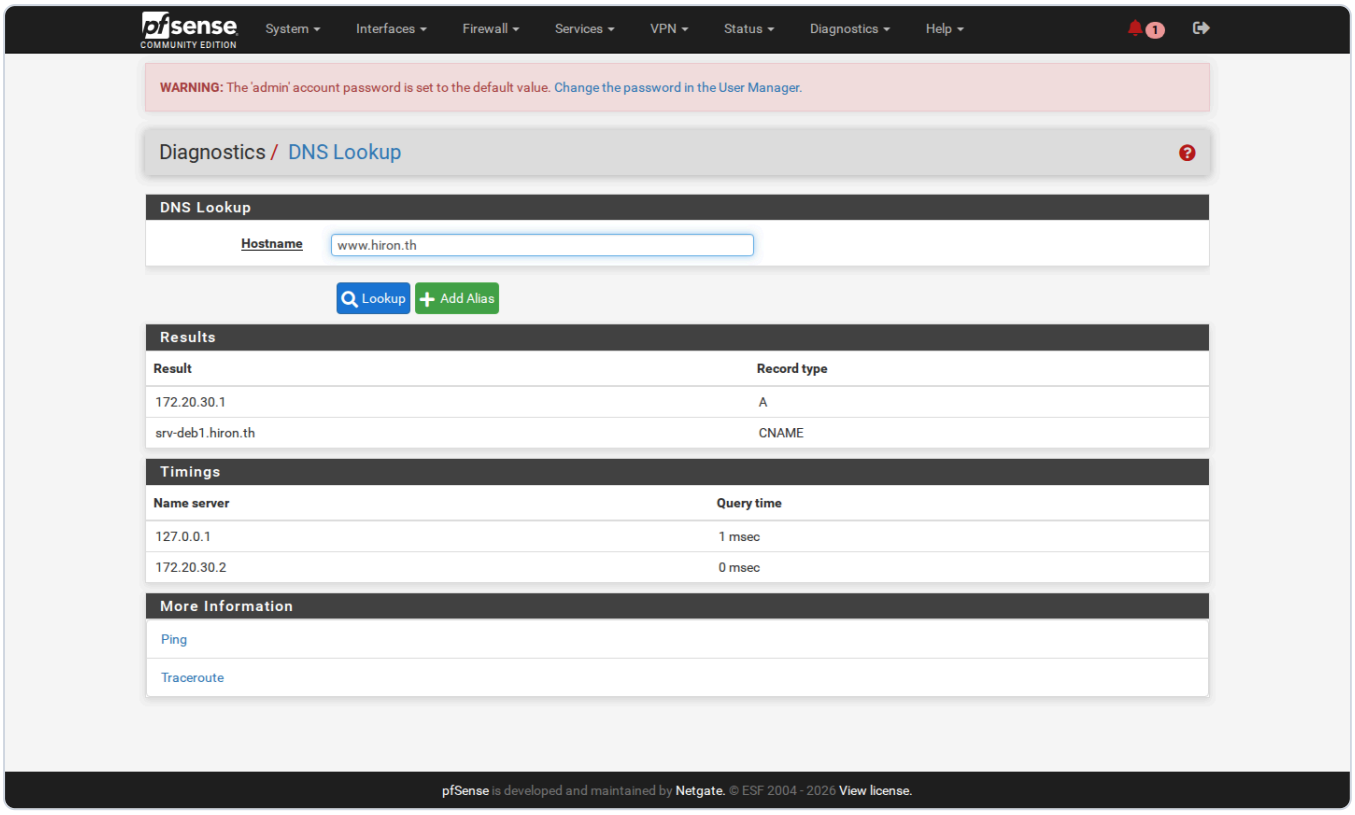
Pour hiron.th et les deux zones inverses, Unbound transmet les requêtes à SRV-DEB2 (172.20.30.2) : équivalent d'un redirecteur conditionnel.


pfSense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Services / DNS Resolver / General Settings
ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend.
General Settings
Advanced Settings
Access Lists
General DNS Resolver Options
Enable
Enable DNS resolver
Listen Port
53
The port used for responding to DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 53.
Enable SSL/TLS Service
Respond to incoming SSL/TLS queries from local clients
Configures the DNS Resolver to act as a DNS over SSL/TLS server which can answer queries from clients which also support DNS over TLS. Activating this option disables automatic interface response routing behavior, thus it works best with specific interface bindings.
SSL/TLS Certificate
GUI default (6ae11e33d6b4f)
The server certificate to use for SSL/TLS service. The CA chain will be determined automatically.
SSL/TLS Listen Port
853
The port used for responding to SSL/TLS DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 853.
Network Interfaces
All
WAN
LAN
WAN IPv6 Link-Local
Interface IP addresses used by the DNS Resolver for responding to queries from clients. If an interface has both IPv4 and IPv6 addresses, both are used. Queries to addresses not selected in this list are discarded. The default behavior is to respond to queries on every available IPv4 and IPv6 address.
Outgoing Network Interfaces
All
WAN
LAN
WAN IPv6 Link-Local
Utilize different network interface(s) that the DNS Resolver will use to send queries to authoritative servers and receive their replies. By default all interfaces are used.
Strict Outgoing Network Interface Binding
Do not send recursive queries if none of the selected Outgoing Network Interfaces are available.
By default the DNS Resolver sends recursive DNS requests over any available interfaces if none of the selected Outgoing Network Interfaces are available. This option makes the DNS Resolver refuse recursive queries.
System Domain Local Zone Type
Transparent
The local-zone type used for the pfSense system domain (System | General Setup | Domain). Transparent is the default.
DNSSEC
Enable DNSSEC Support
Python Module
Enable Python Module
Enable the Python Module.
DNS Query Forwarding
Enable Forwarding Mode
If this option is set, DNS queries will be forwarded to the upstream DNS servers defined under System > General Setup or those obtained via dynamic interfaces such as DHCP, PPP, or OpenVPN (if DNS Server Override is enabled there).
Use SSL/TLS for outgoing DNS Queries to Forwarding Servers
When set in conjunction with DNS Query Forwarding, queries to all upstream forwarding DNS servers will be sent using SSL/TLS on the default port of 853. Note that ALL configured forwarding servers MUST support SSL/TLS queries on port 853.
DHCP Registration
Register DHCP leases in the DNS Resolver
If this option is set, then machines that specify their hostname when requesting an IPv4 DHCP lease will be registered in the DNS Resolver so that their name can be resolved. Note that this will cause the Resolver to reload and flush its resolution cache whenever a DHCP lease is issued. The domain in System > General Setup should also be set to the proper value.
Static DHCP
Register DHCP static mappings in the DNS Resolver
If this option is set, then DHCP static mappings will be registered in the DNS Resolver, so that their name can be resolved. The domain in System > General Setup should also be set to the proper value.
OpenVPN Clients
Register connected OpenVPN clients in the DNS Resolver
If this option is set, then the common name (CN) of connected OpenVPN clients will be registered in the DNS Resolver, so that their name can be resolved. This only works for OpenVPN servers (Remote Access SSL/TLS or User Auth with Username as Common Name option) operating in "tun" mode. The domain in System: General Setup should also be set to the proper value.
Display Custom Options
Display Custom Options
Save
Host Overrides
Host
Parent domain of host
IP to return for host
Description
Actions
Enter any individual hosts for which the resolver's standard DNS lookup process should be overridden and a specific IPv4 or IPv6 address should automatically be returned by the resolver. Standard and also non-standard names and parent domains can be entered, such as 'test', 'nas.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. Any lookup attempt for the host will automatically return the given IP address, and the usual lookup server for the domain will not be queried for the host's records.
Add
Domain Overrides
Domain
Lookup Server IP Address
Description
Actions
hiron.th
172.20.30.2
Zone DNS du TP sur SRV-DEB2



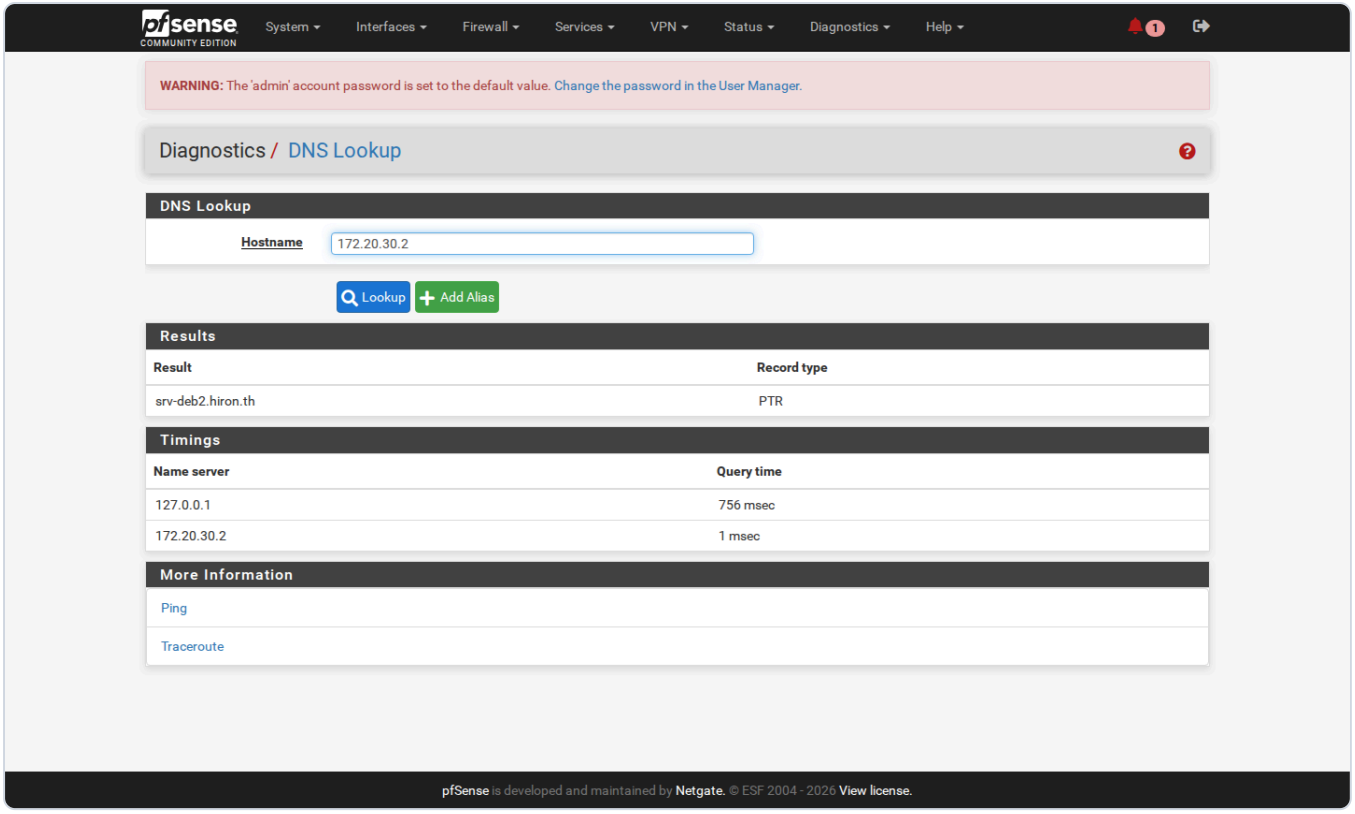
Source : https://172.20.20.2/services_unbound.php
Capturé le 10/11/2026 10:30:53

3.8 — pfSense — DNS Lookup : www.hiron.th résolu



Source : https://172.20.20.2/diag_dns.php?host=www.hiron.th
Capturé le 10/11/2026 10:31:03

3.9 — pfSense — DNS Lookup inverse : 172.20.30.2



Source : https://172.20.20.2/diag_dns.php?host=172.20.30.2
Capturé le 10/11/2026 10:31:15

3.10 — pfSense — vérification en ligne de commande (Unbound + overrides)

Les trois zones (hiron.th et les deux zones inverses) sont transmises à 172.20.30.2 ; les noms Internet restent résolus par Unbound.

```
thomas@pop-os
thomas@pop-os:~$ cat /var/unbound/domainoverrides.conf ; drill @127.0.0.1 <nom> ; drill -x <ip> @127.0.0.1
forward-zone:
        name: "18.172.in-addr.arpa"
        forward-addr: 172.20.30.2
forward-zone:
        name: "30.20.172.in-addr.arpa"
        forward-addr: 172.20.30.2
forward-zone:
        name: "hiron.th"
        forward-addr: 172.20.30.2

www.hiron.th -> 172.20.30.1
pfsense.hiron.th -> 172.20.20.2
172.20.30.1 -> srv-deb1.hiron.th.
172.18.127.254 -> rtux-clt.hiron.th.
debian.org -> 151.101.2.132
```

Capturé le 10/11/2026 10:31:30

Client — tests de résolution

Vérifications avec dig et nslookup depuis le poste client Linux.

4.1 — Client — release/renew et DNS reçu par DHCP

Release/renew avec NetworkManager (nmcli connection down / up). Le DHCP distribue désormais le DNS 172.20.30.2 et le domaine hiron.th.

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ nmcli -t -f IP4.ADDRESS,IP4.GATEWAY,IP4.DNS,IP4.DOMAIN device show ens3; echo '--- /etc/resolv.conf ---'; grep -v '^#' /etc/resolv.conf | sed '/^\s*$/d'
IP4.ADDRESS[1]:172.18.0.1/17
IP4.GATEWAY:172.18.127.254
IP4.DNS[1]:172.20.30.2
IP4.DOMAIN[1]:hiron.th
--- /etc/resolv.conf ---
nameserver 172.20.30.2
search hiron.th
```

Capturé le 10/11/2026 11:03:54

4.2 — Client — enregistrement SOA de la zone (dig + nslookup)

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ dig hiron.th SOA | grep -E 'status|flags|ANSWER SECTION|SOA|SERVER' | grep -v '^;hiron'; echo '---'; nslookup -type=SOA hiron.th
; <<>> DiG 9.18.49-1~deb12u3-Debian <<>> hiron.th SOA
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 42430
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
; EDNS: version: 0, flags:; udp: 1232
;; ANSWER SECTION:
hiron.th.               86400   IN      SOA     srv-deb2.hiron.th. admin.hiron.th. 2026111001 3600 900 604800 86400
;; SERVER: 172.20.30.2#53(172.20.30.2) (UDP)
---
Server:         172.20.30.2
Address:        172.20.30.2#53

hiron.th
        origin = srv-deb2.hiron.th
        mail addr = admin.hiron.th
        serial = 2026111001
        refresh = 3600
        retry = 900
        expire = 604800
        minimum = 86400
```

Capturé le 10/11/2026 11:03:54

4.3 — Client — enregistrement NS de la zone (dig + nslookup)

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ dig hiron.th NS | grep -E 'status|flags|IN\s+(NS|A)\s|SERVER'; echo '---'; nslookup -type=NS hiron.th
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 28906
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2
; EDNS: version: 0, flags:; udp: 1232
hiron.th.               86400   IN      NS      srv-deb2.hiron.th.
srv-deb2.hiron.th.      86400   IN      A       172.20.30.2
;; SERVER: 172.20.30.2#53(172.20.30.2) (UDP)
---
Server:         172.20.30.2
Address:        172.20.30.2#53

hiron.th        nameserver = srv-deb2.hiron.th.
```

Capturé le 10/11/2026 11:03:55

4.4 — Client — résolution des enregistrements créés (directe et inverse)

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ for n in srv-deb1 srv-deb2 rtux-rt rtux-srv rtux-clt pfsense; do printf '%-20s -> %s\n' $n.hiron.th "$(dig $n.hiron.th +short)"; done; echo; dig www.hiron.th +noall +answer; echo; for ip in 172.20.30.1 172.20.30.2 172.20.30.14 172.18.127.254; do printf '%-16s -> %s\n' $ip "$(dig -x $ip +short)"; done; echo; nslookup www.hiron.th | tail -n +3; nslookup 172.18.127.254 | grep name
srv-deb1.hiron.th    -> 172.20.30.1
srv-deb2.hiron.th    -> 172.20.30.2
rtux-rt.hiron.th     -> 172.20.20.6
rtux-srv.hiron.th    -> 172.20.30.14
rtux-clt.hiron.th    -> 172.18.127.254
pfsense.hiron.th     -> 172.20.20.2

www.hiron.th.           86400   IN      CNAME   srv-deb1.hiron.th.
srv-deb1.hiron.th.      86400   IN      A       172.20.30.1

172.20.30.1      -> srv-deb1.hiron.th.
172.20.30.2      -> srv-deb2.hiron.th.
172.20.30.14     -> rtux-srv.hiron.th.
172.18.127.254   -> rtux-clt.hiron.th.


www.hiron.th    canonical name = srv-deb1.hiron.th.
Name:   srv-deb1.hiron.th
Address: 172.20.30.1

254.127.18.172.in-addr.arpa     name = rtux-clt.hiron.th.
```

Capturé le 10/11/2026 11:03:55

4.5 — Client — connectivité par FQDN puis par nom d'hôte

Le nom court est complété grâce au suffixe de recherche hiron.th reçu par DHCP.

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ ping -c 2 srv-deb1.hiron.th; ping -c 2 srv-deb2.hiron.th; ping -c 2 srv-deb1; ping -c 2 srv-deb2
PING srv-deb1.hiron.th (172.20.30.1) 56(84) bytes of data.
64 bytes from srv-deb1.hiron.th (172.20.30.1): icmp_seq=1 ttl=63 time=0.510 ms
64 bytes from srv-deb1.hiron.th (172.20.30.1): icmp_seq=2 ttl=63 time=1.37 ms

--- srv-deb1.hiron.th ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.510/0.942/1.374/0.432 ms
PING srv-deb2.hiron.th (172.20.30.2) 56(84) bytes of data.
64 bytes from srv-deb2.hiron.th (172.20.30.2): icmp_seq=1 ttl=63 time=1.19 ms
64 bytes from srv-deb2.hiron.th (172.20.30.2): icmp_seq=2 ttl=63 time=1.57 ms

--- srv-deb2.hiron.th ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.190/1.378/1.567/0.188 ms
PING srv-deb1.hiron.th (172.20.30.1) 56(84) bytes of data.
64 bytes from srv-deb1.hiron.th (172.20.30.1): icmp_seq=1 ttl=63 time=1.17 ms
64 bytes from srv-deb1.hiron.th (172.20.30.1): icmp_seq=2 ttl=63 time=1.01 ms

--- srv-deb1.hiron.th ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.013/1.089/1.165/0.076 ms
PING srv-deb2.hiron.th (172.20.30.2) 56(84) bytes of data.
64 bytes from srv-deb2.hiron.th (172.20.30.2): icmp_seq=1 ttl=63 time=1.21 ms
64 bytes from srv-deb2.hiron.th (172.20.30.2): icmp_seq=2 ttl=63 time=0.910 ms

--- srv-deb2.hiron.th ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.910/1.061/1.212/0.151 ms
```

Capturé le 10/11/2026 11:04:00

DNS secondaire sur SRV-DEB1

SRV-DEB1 reçoit une copie des zones par transfert (AXFR) et répond si SRV-DEB2 tombe.

5.1 — SRV-DEB2 — transfert de zone autorisé vers SRV-DEB1 (allow-transfer, also-notify)

Seul SRV-DEB1 (172.20.30.1) peut copier les zones (AXFR) ; also-notify le prévient à chaque modification.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ hostname; grep -v '^//' /etc/bind/named.conf.local | sed '/^\s*$/d'; sudo named-checkconf
&& echo 'named-checkconf : OK'
SRV-DEB2
zone "hiron.th" {
        type master;
        file "/etc/bind/db.hiron.th";
        allow-transfer { 172.20.30.1; };
        also-notify { 172.20.30.1; };
};
zone "30.20.172.in-addr.arpa" {
        type master;
        file "/etc/bind/db.172.20.30";
        allow-transfer { 172.20.30.1; };
        also-notify { 172.20.30.1; };
};
zone "18.172.in-addr.arpa" {
        type master;
        file "/etc/bind/db.172.18";
        allow-transfer { 172.20.30.1; };
        also-notify { 172.20.30.1; };
};
named-checkconf : OK
```

Capturé le 10/11/2026 11:20:06

5.2 — SRV-DEB2 — SRV-DEB1 déclaré en NS, serial 2026111002

Le serial augmente à chaque modification : c'est lui que le secondaire compare pour savoir s'il doit recopier la zone.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ for z in 'hiron.th db.hiron.th' '30.20.172.in-addr.arpa db.172.20.30' '18.172.in-addr.arp
a db.172.18'; do set -- $z; echo "== $2"; sudo grep -E 'IN\s+NS' /etc/bind/$2; sudo named-checkzone $1 /et
c/bind/$2 | tr '\n' ' '; echo; done; systemctl is-active named; echo '--- NS de la zone ---'; dig @127.0.
0.1 hiron.th NS +short; dig @127.0.0.1 hiron.th SOA +short
== db.hiron.th
@               IN      NS      srv-deb2.hiron.th.
@               IN      NS      srv-deb1.hiron.th.
zone hiron.th/IN: loaded serial 2026111002 OK
== db.172.20.30
@       IN      NS      srv-deb2.hiron.th.
@       IN      NS      srv-deb1.hiron.th.
zone 30.20.172.in-addr.arpa/IN: loaded serial 2026111002 OK
== db.172.18
@       IN      NS      srv-deb2.hiron.th.
@       IN      NS      srv-deb1.hiron.th.
zone 18.172.in-addr.arpa/IN: loaded serial 2026111002 OK
active
--- NS de la zone ---
srv-deb1.hiron.th.
srv-deb2.hiron.th.
srv-deb2.hiron.th. admin.hiron.th. 2026111002 3600 900 604800 86400
```

Capturé le 10/11/2026 11:20:07

5.3 — DHCP KEA — DNS principal + secondaire distribués (sur les deux serveurs)

Ordre : 172.20.30.2 (principal) d'abord, 172.20.30.1 (secondaire) si le premier ne répond pas.

thomas@pop-os

```
thomas@pop-os:~$ grep domain-name-servers /etc/kea/kea-dhcp4.conf ; kea-dhcp4 -t ; systemctl is-active kea
-dhcp4-server  (SRV-DEB2 puis SRV-DEB1)
## SRV-DEB2
29:        { "name": "domain-name-servers", "data": "172.20.30.2, 172.20.30.1" }
kea-dhcp4 -t : configuration valide
active
## SRV-DEB1
29:        { "name": "domain-name-servers", "data": "172.20.30.2, 172.20.30.1" }
kea-dhcp4 -t : configuration valide
active
```

Capturé le 10/11/2026 11:25:16

5.4 — SRV-DEB1 — installation de BIND9 et options (redirecteur 1.1.1.1)

Même redirecteur et mêmes droits que le principal : si SRV-DEB2 tombe, SRV-DEB1 doit aussi résoudre Internet pour les clients.

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ hostname; dpkg -l bind9 | grep ^ii | awk '{print $1, $2, $3}'; systemctl is-active named;
grep -E 'forwarders|1.1.1.1|forward only|allow-|acl' /etc/bind/named.conf.options
SRV-DEB1
ii bind9 1:9.18.49-1~deb12u3
active
acl "reseaux_tp" {
        forwarders {
                1.1.1.1;
        forward only;
        allow-query { reseaux_tp; };
        allow-recursion { reseaux_tp; };
```

Capturé le 10/11/2026 11:27:49

5.5 — SRV-DEB1 — zones secondaires (type slave)

Les copies sont écrites par BIND dans /var/cache/bind (l'utilisateur bind ne peut pas écrire dans /etc/bind).

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ grep -v '^//' /etc/bind/named.conf.local | sed '/^\s*$/d'; sudo named-checkconf && echo
'named-checkconf : OK'
zone "hiron.th" {
        type slave;
        masters { 172.20.30.2; };
        file "/var/cache/bind/db.hiron.th";
};
zone "30.20.172.in-addr.arpa" {
        type slave;
        masters { 172.20.30.2; };
        file "/var/cache/bind/db.172.20.30";
};
zone "18.172.in-addr.arpa" {
        type slave;
        masters { 172.20.30.2; };
        file "/var/cache/bind/db.172.18";
};
named-checkconf : OK
```

Capturé le 10/11/2026 11:27:49

5.6 — SRV-DEB1 — transfert de zone (AXFR) depuis SRV-DEB2

Les 3 zones ont été copiées automatiquement ; le serial (2026111002) est identique à celui du principal.

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ ls -l /var/cache/bind/db.*; sudo journalctl -u named --no-pager | grep -E 'Transfer compl
eted|transferred serial' | tail -6; echo '--- SOA sur le secondaire ---'; dig @127.0.0.1 hiron.th SOA +sho
rt; dig @127.0.0.1 www.hiron.th +short; dig @127.0.0.1 -x 172.20.30.2 +short; dig @127.0.0.1 -x 172.18.12
7.254 +short
-rw-r--r-- 1 bind bind 275 10 nov.  11:27 /var/cache/bind/db.172.18
-rw-r--r-- 1 bind bind 413 10 nov.  11:27 /var/cache/bind/db.172.20.30
-rw-r--r-- 1 bind bind 506 10 nov.  11:27 /var/cache/bind/db.hiron.th
nov. 10 11:27:11 SRV-DEB1 named[1752]: zone hiron.th/IN: transferred serial 2026111002
nov. 10 11:27:11 SRV-DEB1 named[1752]: transfer of 'hiron.th/IN' from 172.20.30.2#53: Transfer completed:
1 messages, 11 records, 298 bytes, 0.001 secs (298000 bytes/sec) (serial 2026111002)
nov. 10 11:27:11 SRV-DEB1 named[1752]: zone 30.20.172.in-addr.arpa/IN: transferred serial 2026111002
nov. 10 11:27:11 SRV-DEB1 named[1752]: transfer of '30.20.172.in-addr.arpa/IN' from 172.20.30.2#53: Transf
er completed: 1 messages, 7 records, 230 bytes, 0.004 secs (57500 bytes/sec) (serial 2026111002)
nov. 10 11:27:11 SRV-DEB1 named[1752]: zone 18.172.in-addr.arpa/IN: transferred serial 2026111002
nov. 10 11:27:11 SRV-DEB1 named[1752]: transfer of '18.172.in-addr.arpa/IN' from 172.20.30.2#53: Transfer
completed: 1 messages, 5 records, 219 bytes, 0.004 secs (54750 bytes/sec) (serial 2026111002)
--- SOA sur le secondaire ---
srv-deb2.hiron.th. admin.hiron.th. 2026111002 3600 900 604800 86400
srv-deb1.hiron.th.
172.20.30.1
srv-deb2.hiron.th.
rtux-clt.hiron.th.
```

Capturé le 10/11/2026 11:27:50

Basculement — panne du DNS principal

On arrête BIND sur SRV-DEB2 et on vérifie avec Wireshark que le client interroge le secondaire.

6.1 — Client — release/renew : deux DNS reçus par DHCP

Release/renew par NetworkManager (nmcli connection down / up), équivalent de dhclient -r / dhclient.

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ nmcli -f IP4.ADDRESS,IP4.DNS,IP4.DOMAIN device show ens3; echo '--- /etc/resolv.conf ---';
grep -v '^#' /etc/resolv.conf | sed '/^\s*$/d'
IP4.ADDRESS[1]:                         172.18.0.1/17
IP4.DNS[1]:                             172.20.30.2
IP4.DNS[2]:                             172.20.30.1
IP4.DOMAIN[1]:                          hiron.th
--- /etc/resolv.conf ---
nameserver 172.20.30.2
nameserver 172.20.30.1
search hiron.th
```

Capturé le 10/11/2026 11:30:42

6.2 — Client — les deux serveurs DNS répondent

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ echo '== principal SRV-DEB2'; dig @172.20.30.2 www.hiron.th +short; dig @172.20.30.2 hiro
n.th SOA +short; echo '== secondaire SRV-DEB1'; dig @172.20.30.1 www.hiron.th +short; dig @172.20.30.1 hir
on.th SOA +short
== principal SRV-DEB2
srv-deb1.hiron.th.
172.20.30.1
srv-deb2.hiron.th. admin.hiron.th. 2026111002 3600 900 604800 86400
== secondaire SRV-DEB1
srv-deb1.hiron.th.
172.20.30.1
srv-deb2.hiron.th. admin.hiron.th. 2026111002 3600 900 604800 86400
```

Capturé le 10/11/2026 11:30:42

7.1 — SRV-DEB2 — arrêt du service BIND9 (panne simulée)

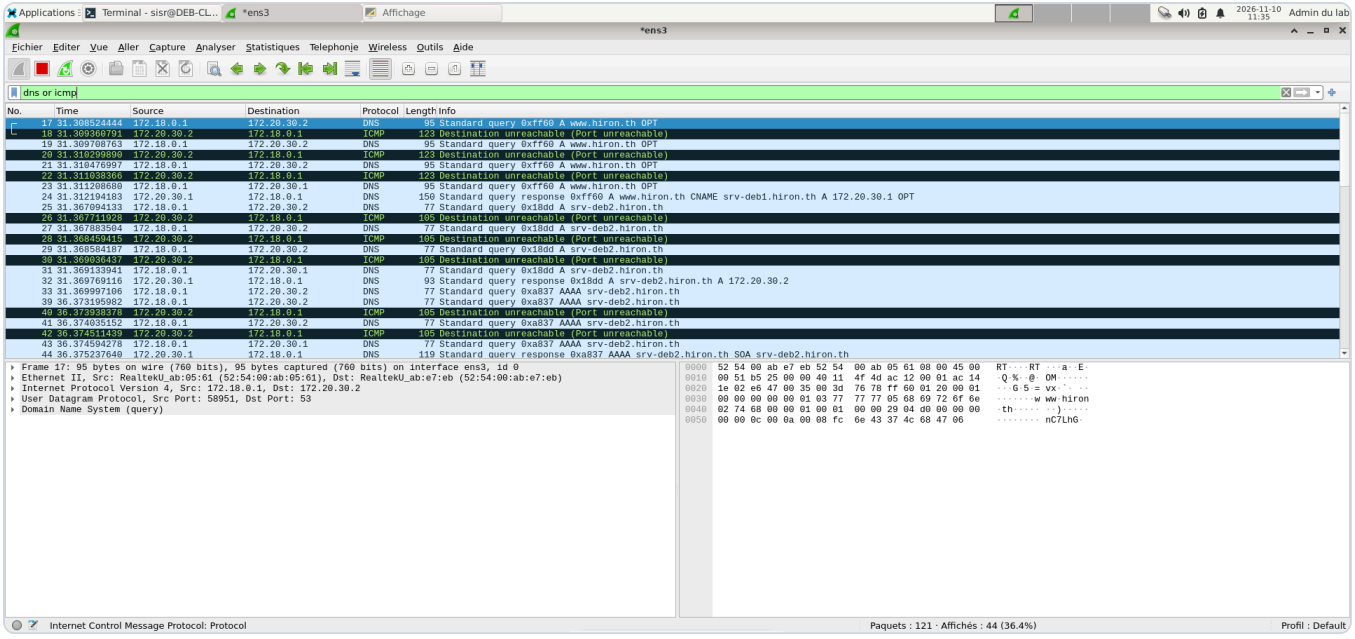
Le serveur reste allumé, seul le service DNS est arrêté.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ hostname; systemctl status named --no-pager | head -5
SRV-DEB2
○ named.service - BIND Domain Name Server
     Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: inactive (dead) since Tue 2026-11-10 11:31:59 CET; 3min 13s ago
   Duration: 12min 33.578s
       Docs: man:named(8)
```

Capturé le 10/11/2026 11:35:13

7.2 — Client — Wireshark (filtre DNS) pendant la panne du principal



Source : écran de la VM lab-deb-clt

Capturé le 10/11/2026 11:35:13

7.3 — Client — résolution assurée par le DNS secondaire

Le principal étant arrêté, la réponse vient de SERVER 172.20.30.1 (SRV-DEB1).

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ dig www.hiron.th | grep -E 'status|ANSWER SECTION|IN\s+(A|CNAME)\s|SERVER|connection refused'; echo '---'; nslookup srv-deb2.hiron.th | grep -E 'Server|Name|Address: 172'; echo '---'; ping -c 2 srv-deb1 | grep -E 'PING|packets'
;; communications error to 172.20.30.2#53: connection refused
;; communications error to 172.20.30.2#53: connection refused
;; communications error to 172.20.30.2#53: connection refused
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 7916
;; ANSWER SECTION:
www.hiron.th.           86400   IN      CNAME   srv-deb1.hiron.th.
srv-deb1.hiron.th.      86400   IN      A       172.20.30.1
;; SERVER: 172.20.30.1#53(172.20.30.1) (UDP)
---
Server:         172.20.30.1
Name:   srv-deb2.hiron.th
Address: 172.20.30.2
---
PING srv-deb1.hiron.th (172.20.30.1) 56(84) bytes of data.
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
```

Capturé le 10/11/2026 11:35:31

Mise à jour dynamique DHCP → DNS

Le serveur DHCP inscrit lui-même les clients dans la zone (DDNS, clé TSIG).

8.1 — SRV-DEB2 — redémarrage du service BIND9

Fin de la panne simulée : le DNS principal répond de nouveau.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ hostname; systemctl status named --no-pager | head -5; dig @127.0.0.1 www.hiron.th +short
SRV-DEB2
● named.service - BIND Domain Name Server
     Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 11:40:08 CET; 40s ago
       Docs: man:named(8)
   Main PID: 2765 (named)
srv-deb1.hiron.th.
172.20.30.1
```

Capturé le 10/11/2026 11:40:49

8.2 — SRV-DEB2 — clé TSIG et zones dynamiques dans /var/lib/bind

Clé générée par tsig-keygen -a hmac-sha256 ddns-key (secret masqué). AppArmor n'autorise BIND à écrire que dans /var/lib/bind : les zones mises à jour dynamiquement y sont déplacées.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ ls -l /etc/bind/ddns.key /var/lib/bind/; sudo sed -E 's/(secret ").*(";)/\1••••••••\2/' /
etc/bind/ddns.key
-rw-r----- 1 root bind  100 10 nov.  11:37 /etc/bind/ddns.key

/var/lib/bind/:
total 8
-rw-r--r-- 1 bind bind 261 10 nov.  11:18 db.172.18
-rw-r--r-- 1 bind bind 864 10 nov.  11:17 db.hiron.th
key "ddns-key" {
        algorithm hmac-sha256;
        secret "••••••••";
};
```

Capturé le 10/11/2026 11:40:49

8.3 — SRV-DEB2 — allow-update par clé TSIG (named.conf.local)

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ grep -v '^//' /etc/bind/named.conf.local | sed '/^\s*$/d'; sudo named-checkconf && echo 'named-checkconf : OK'; systemctl is-active named; sudo journalctl -u named --no-pager --since '-10min' | grep -iE 'loaded serial|denied|error' | tail -5
include "/etc/bind/ddns.key";
zone "hiron.th" {
        type master;
        file "/var/lib/bind/db.hiron.th";
        allow-transfer { 172.20.30.1; };
        also-notify { 172.20.30.1; };
        allow-update { key ddns-key; };
};
zone "30.20.172.in-addr.arpa" {
        type master;
        file "/etc/bind/db.172.20.30";
        allow-transfer { 172.20.30.1; };
        also-notify { 172.20.30.1; };
};
zone "18.172.in-addr.arpa" {
        type master;
        file "/var/lib/bind/db.172.18";
        allow-transfer { 172.20.30.1; };
        also-notify { 172.20.30.1; };
        allow-update { key ddns-key; };
};
named-checkconf : OK
active
nov. 10 11:40:08 SRV-DEB2 named[2765]: zone 18.172.in-addr.arpa/IN: loaded serial 2026111002
nov. 10 11:40:08 SRV-DEB2 named[2765]: zone 30.20.172.in-addr.arpa/IN: loaded serial 2026111002
nov. 10 11:40:08 SRV-DEB2 named[2765]: zone 255.in-addr.arpa/IN: loaded serial 1
nov. 10 11:40:08 SRV-DEB2 named[2765]: zone localhost/IN: loaded serial 2
nov. 10 11:40:08 SRV-DEB2 named[2765]: zone hiron.th/IN: loaded serial 2026111002
```

Capturé le 10/11/2026 11:40:49

8.4 — KEA D2 (kea-dhcp-ddns) — configuration TSIG et zones à mettre à jour

D2 signe chaque DNS UPDATE avec la clé TSIG ddns-key (même secret que /etc/bind/ddns.key, masqué ici) et l'envoie à SRV-DEB2 pour hiron.th et 18.172.in-addr.arpa. Configuration identique sur SRV-DEB1.

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ sudo sed -E 's/("secret": ").*(")/\1••••••••\2/' /etc/kea/kea-dhcp-ddns.conf | grep -v '^
\s*$'
// Kea DHCP-DDNS (D2) - TP DNS - met à jour BIND (SRV-DEB2) à chaque bail
{
"DhcpDdns": {
    "ip-address": "127.0.0.1",
    "port": 53001,
    "control-socket": {
        "socket-type": "unix",
        "socket-name": "/run/kea/kea-ddns-ctrl-socket"
    },
    // Clé TSIG : identique à /etc/bind/ddns.key sur SRV-DEB2
    "tsig-keys": [
        {
            "name": "ddns-key",
            "algorithm": "HMAC-SHA256",
            "secret": "••••••••"
        }
    ],
    // Zone directe : enregistrements A des clients
    "forward-ddns": {
        "ddns-domains": [
            {
                "name": "hiron.th.",
                "key-name": "ddns-key",
                "dns-servers": [ { "ip-address": "172.20.30.2" } ]
            }
        ]
    },
    // Zone inverse LAN Clients : enregistrements PTR
    "reverse-ddns": {
        "ddns-domains": [
            {
                "name": "18.172.in-addr.arpa.",
                "key-name": "ddns-key",
                "dns-servers": [ { "ip-address": "172.20.30.2" } ]
            }
        ]
    },
    "loggers": [
        {
            "name": "kea-dhcp-ddns",
            "output_options": [ { "output": "stdout" } ],
            "severity": "INFO"
        }
    ]
}
}
```

Capturé le 10/11/2026 11:51:48

8.5 — KEA DHCPv4 — activation du DNS dynamique (sur les deux serveurs)

thomas@pop-os

```
thomas@pop-os:~$ grep ddns /etc/kea/kea-dhcp4.conf ; kea-dhcp-ddns -t ; kea-dhcp4 -t ; systemctl is-active
(SRV-DEB2 puis SRV-DEB1)
## SRV-DEB2
    "dhcp-ddns": { "enable-updates": true },
    "ddns-send-updates": true,
    "ddns-qualifying-suffix": "hiron.th.",
    "ddns-override-client-update": true,
    "ddns-override-no-update": true,
    "ddns-replace-client-name": "when-not-present",
    "ddns-generated-prefix": "client",
    "ddns-update-on-renew": true,
kea-dhcp-ddns -t : configuration valide
kea-dhcp4 -t : configuration valide
services : active active
## SRV-DEB1
    "dhcp-ddns": { "enable-updates": true },
    "ddns-send-updates": true,
    "ddns-qualifying-suffix": "hiron.th.",
    "ddns-override-client-update": true,
    "ddns-override-no-update": true,
    "ddns-replace-client-name": "when-not-present",
    "ddns-generated-prefix": "client",
    "ddns-update-on-renew": true,
kea-dhcp-ddns -t : configuration valide
kea-dhcp4 -t : configuration valide
services : active active
```

Capturé le 10/11/2026 11:51:48

8.6 — Client — enregistrement automatique dans le DNS (DDNS)

Après un release/renew, le client est inscrit tout seul (A + PTR) sur le principal, puis recopié sur le secondaire (NOTIFY + transfert). BIND a incrémenté le serial lui-même.

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ hostname; ip -4 -br a show ens3; echo '--- principal (SRV-DEB2) ---'; dig deb-clt.hiron.th
+short; dig -x 172.18.0.1 +short; echo '--- secondaire (SRV-DEB1) ---'; dig @172.20.30.1 deb-clt.hiron.th
+short; dig @172.20.30.1 -x 172.18.0.1 +short; echo '--- SOA (serial incrémenté par BIND) ---'; dig hiron.
th SOA +short; dig @172.20.30.1 hiron.th SOA +short
DEB-CLT
ens3             UP             172.18.0.1/17
--- principal (SRV-DEB2) ---
172.18.0.1
deb-clt.hiron.th.
--- secondaire (SRV-DEB1) ---
172.18.0.1
deb-clt.hiron.th.
--- SOA (serial incrémenté par BIND) ---
srv-deb2.hiron.th. admin.hiron.th. 2026111003 3600 900 604800 86400
srv-deb2.hiron.th. admin.hiron.th. 2026111003 3600 900 604800 86400
```

Capturé le 10/11/2026 11:55:50

8.7 — Journaux — KEA D2 et BIND : mise à jour signée acceptée

BIND vérifie la signature TSIG (signer "ddns-key" approved) avant d'ajouter les enregistrements A et PTR, puis notifie le secondaire qui recopie la zone (IXFR).

thomas@pop-os

```
thomas@pop-os:~$ journalctl -u kea-dhcp-ddns-server ; journalctl -u named  (SRV-DEB1 / SRV-DEB2)
== KEA D2 (SRV-DEB1, DHCP principal) ==
DHCP_DDNS_ADD_SUCCEEDED DHCP_DDNS Request ID 000101D9178927B1559164027C8E923DDFFF24C81CBBCC14FF82477A16D45
90BDF9ADD: suc
== BIND (SRV-DEB2) ==
172.20.30.1#33891/key ddns-key: signer "ddns-key" approved
172.20.30.1#33891/key ddns-key: updating zone 'hiron.th/IN': adding an RR at 'deb-clt.hiron.th' A 172.18.
0.1
zone hiron.th/IN: sending notifies (serial 2026111003)
172.20.30.1#37484/key ddns-key: signer "ddns-key" approved
172.20.30.1#37484/key ddns-key: updating zone '18.172.in-addr.arpa/IN': adding an RR at '1.0.18.172.in-add
r.arpa' PTR deb-clt.hiron.th.
zone 18.172.in-addr.arpa/IN: sending notifies (serial 2026111003)
== Transfert vers le secondaire (SRV-DEB1) ==
zone hiron.th/IN: transferred serial 2026111003
zone 18.172.in-addr.arpa/IN: transferred serial 2026111003
```

Capturé le 10/11/2026 11:56:11

8.8 — Bail KEA — indicateurs DDNS (fqdn_fwd / fqdn_rev)

fqdn_fwd = 1 et fqdn_rev = 1 : les enregistrements A et PTR ont été créés pour ce bail (nom deb-clt.hiron.th).

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ head -1 /var/lib/kea/kea-leases4.csv | cut -d, -f1,2,7,8,9; sudo grep 172.18.0.1 /var/li
b/kea/kea-leases4.csv | tail -1 | cut -d, -f1,2,7,8,9
address,hwaddr,fqdn_fwd,fqdn_rev,hostname
172.18.0.1,52:54:00:ab:05:61,1,1,deb-clt.hiron.th
```

Capturé le 10/11/2026 11:56:11

Synthèse

9.1 — Synthèse et incidents rencontrés

Objectifs atteints : DNS principal BIND9 sur SRV-DEB2 (zone hiron.th, redirecteur 1.1.1.1, zones inverses), intégration DHCP KEA, RTUX, SRV-DEB1 et pfSense, DNS secondaire sur SRV-DEB1 avec transfert de zone (AXFR/IXFR + NOTIFY), basculement vérifié avec Wireshark, mise à jour dynamique DHCP → DNS signée TSIG. Incidents : (1) pfSense : NXDOMAIN renvoyé par le résolveur local Unbound, corrigé par des Domain Overrides ; (2) fichiers modifiés par erreur sur SRV-DEB1 au lieu de SRV-DEB2, sans conséquence (bind9 absent, rien d'enregistré) ; (3) option DNS de KEA non enregistrée sur SRV-DEB1 au premier essai ; (4) texte collé en trop dans kea-dhcp-ddns.conf sur SRV-DEB1, empêchant le redémarrage de KEA (DDNS désactivé) : fichier corrigé et service redémarré.

Capturé le 10/11/2026 11:56:28